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The Evolution of Intelligent Automation as a Sourcing Option

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Introduction

This chapter looks at the evolution of automation as a service option and is based on a much revised and updated interview conducted by Gabe Piccoli, then *MIS Quarterly Executive Editor* in Chief. He initially asked Mary Lacity and Leslie Willcocks to discuss their six-year research program, with the goal of consolidating what is known about Robotic Process Automation (RPA) and Cognitive Automation (CA) which, in

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combination are often today called Intelligent Automation (IA) and even included as Artificial Intelligence (AI). The chapter has been extended to cover also the 2021–2023 period to take into account more recent consolidating research. The authors also identify the challenges and ways forward for digital systems leaders, and a research agenda for academics in this field.

By 2023 the hype levels around AI and its impacts were higher than ever before. This chapter uses evidence from organizations to provide a more sober and complex picture than that feeding into the popular imagination.

The Terminology Challenge

We are going to reflect on the overall findings and most impactful lessons from the authors' nine-year research program. It is useful to start by dealing with the confusing terminology used in the automation field.

The research focuses on how enterprises automate services using a variety of digital technologies. When we began studying this space back in 2014, we encountered a dizzying array of automation products marketed as scripting tools, software robots, robotic process automation, artificial intelligence, desktop automation, cognitive computing, process mining, business process management automation and machine learning, to name but a few. The market was very confusing to practitioners, academics and to us.

To make sense of the space, we looked at how these tools worked, the type of data used as input, how they processed data and the type of results produced. From this work, we identified a continuum of automation tools, with the realm of robotic process automation (RPA) at one end and the realm of cognitive automation (CA) at the other. Many now commonly call the latter “artificial intelligence” (AI).

The realm of RPA consists of tools that automate tasks that have clearly defined rules for processing structured data to produce deterministic outcomes. For example, a “software robot” can be configured to process tasks the way humans do, by giving it a logon ID, password and playbook for executing processes. RPA tools are ideally suited for

automating those mindless “swivel chair” chores performed by humans, like taking structured data from spreadsheets and applying some rules to update an ERP system. We had the early insight that RPA tools “*take the robot out of the human*,” meaning that the tedious parts of a person’s job could be automated, leaving the human to do more interesting work that requires judgment and social skills. This insight resonated so much with practitioners, that we had to chase down more than one supplier and conference organizer who adopted the phrase as their marketing slogan without crediting us. The top RPA providers by market share as of 2023 were Automation Anywhere, Blue Prism and UiPath.

It is important to note that, because of the consequences for ease of scaling, these software providers do not all provide the same thing. There are variants of RPA ranging from desktop-assisted RPA, through enterprise RPA, self-development packages to cloud-based services.

The realm of cognitive automation consists of more powerful software suites that automate or augment tasks that do not have clearly defined rules. We do not like to call such software “artificial intelligence” because we believe the AI label over-states what these tools do. With CA technologies, inference-based algorithms process data to produce probabilistic outcomes. The realm of CA includes a variety of tools, such as tools that analyze data based on supervised and unsupervised machine learning, and deep learning algorithms. Some of the algorithms have been around for decades, but the computational power and memory needed to execute them on big data have only recently become available.

The input data for CA tools is often unstructured, such as free-form text, either written or spoken. For example, at Deakin University, CA was used to answer natural language inquiries from students. However, the input data can also be highly structured, such as the pixels in an image. Google’s Machine Learning Kit, IPsoft’s Amelia, IBM’s Watson suite and Expert Systems’ Cogito are examples of CA tools, likewise ChatGTP and similar software suites based on large language models (LLM—a type of artificial intelligence (AI) algorithm that uses deep learning techniques and massively large data sets to understand, summarize, generate and predict new content).

People refer to “strong” AI as using computers to do what human minds can do.¹ The vast majority of organizations are a long way off that! In our view, however, “AI” is widely and misleadingly used, especially by vendors, as an umbrella terms for RPA and CA, as well as for much more advanced software that has not even made it out of the laboratory. The claims by vendors are not helped by the misleading metaphor of comparing the human brain with computing. Most “AI applications” in businesses today can be described as “*weak, weak AI*”—algorithms driven by massive computing power. This is not “intelligence”; it is machine learning—we call it “statistics on steroids” (Fersht, 2012; Willcocks, 2020b).

Though the realm of CA is vast, our research examined how the tools were used to automate or augment back-office processes and customer-facing services, in line with our early interest in RPA. Typical examples we witnessed included processing medical claims, answering customer queries and categorizing user requests to route them to the humans who could help.

The Recent Evolution of Automation

A quick history of RPA is useful at this point. In 2012, Phil Fersht, founder of the outsourcing consulting firm Horses for Sources (HFS), used the term RPA in a provocative report entitled *Greetings from Robo-tistan, outsourcing’s cheapest new destination*. This is a nice idea as it encapsulates automation’s ability to replace human labor—the standard selling point of outsourcing vendors—by cheaper, quicker, more accurate machines. It highlighted a UK-based start-up called Blue Prism, which was founded in 2001. Blue Prism did not become well-known until its chief marketing officer, Patrick Geary, started calling its product “robotic process automation” sometime in 2012. That term really resonated with practitioners, so much so, that other automation companies started

¹ For a more comprehensive coverage of the types of strong and weak AI, see Benbya, H., Davenport, T. H. and Pachidi, S. (2020), “Special Issue Editorial: Artificial Intelligence in Organizations: Current State and Future Opportunities,” *MIS Quarterly Executive* (19:4), December, available at: <https://aisel.aisnet.org/misq/vol19/iss4/4>.

rebranding their tools as RPA. By 2016, there were over two dozen companies saying they provided RPA tools, with a claimed market size of \$600 million (Fersht & Snowdon, 2018).

There was a desperate need for RPA standards, so Lee Coulter, then CEO of Ascension Shared Services, started an initiative at IEEE, and in December 2016 became the chair of the IEEE Working Group on Standards in Intelligent Process Automation. The group published the first standard in 2017, which distinguished between enterprise RPA designed for an organization and robotic desktop automation (RDA) designed for a single desktop user (IEEE, 2017). Blue Prism began as an RPA provider; Automation Anywhere began as an RDA provider. Both companies' products have evolved into more sophisticated platforms that include natural language processing and machine-learning features.

The RPA market was between \$US2 billion and \$US4 billion in 2020, depending on which consulting report you read.² Nearly every source predicts the annual growth rate will be between 30 and 50% for the foreseeable future.³ C-suite priorities for emerging technologies have shifted rapidly because of Covid-19. Though the pandemic prompted many enterprises to postpone horizon technologies like edge computing and blockchains, they became laser-focused on technologies that produce rapid return-on-investments (ROIs), and the top of that list was process automation (KPMG, 2020). However, RPA has become a quite small proportion of the overall combined market for RPA, Intelligent Automation and "AI." On a Research and Markets estimate the global artificial intelligence market size was US\$136.55 billion in 2022 and expected to reach US\$1,812 billion by 2030, expanding at a CAGR of 37.3 percent from 2023 to 2030.

² See: (1) Robotic process automation (RPA) market revenues worldwide from 2017 to 2023, Statista, January 2020, available at <https://www.statista.com/statistics/740440/worldwide-robotic-process-automation-market-size/>; and (2) Fersht and Snowdon (2018).

³ Allied Market Research, reporting on 13 October 2020, forecast that the RPA market at end of 2027 will be \$19.53 billion—an annual compound growth rate of 36.4%. A Deloitte survey reporting in December 2020 predicted the RPA market would grow at 40.6% per year to reach \$25.6 billion by 2027.

Emerging Challenges

Organizations find it difficult to adapt CA tools designed for a specific context like chess, Jeopardy or Go for use in other contexts like processing health records, mortgage applications and calls to helpdesks. The early enterprise adopters experienced painful and expensive implementations, mostly due to the data challenge. Our case companies adopted CA tools with supervised machine learning, which needs thousands of labeled data examples to enable the machine-learning algorithms to reach an acceptable level of proficiency. However, as much as 80% of an organization's data is "dark," meaning that the data is un-locatable, untapped or untagged. Enterprises adopters of CA tools first had to create new data and clean up dirty data that was missing, duplicated, incorrect, inconsistent or outdated. They also struggled with "difficult data," which we define as accurate and valid data that is hard for a machine to read—for example, a fuzzy image, unexpected data types or sophisticated natural language text. In our CA cases, much of the work to sort out data problems was done by tedious human review.

But we did find examples of organizations that eventually got value from their CA adoptions, including Deakin University, Zurich Insurance and KPMG. As best practices emerge, more organizations are finding success, though CA implementations still tend to be islands of automation rather than enterprise-wide deployments. The size of the CA market in 2020 was reported to be somewhere between \$US50 billion and \$US150 billion.⁴

RPA and CA have different histories that, from around 2018, have been now converging to what practitioners are calling "intelligent automation." The idea of intelligent automation is to institutionalize a well-designed automation program using a platform for pluggable tools that are best-in-class. In our case studies, business value was not derived

⁴ See, for example: (1) *Artificial Intelligence Market Size, Share & Trends Analysis Report, Grand View Research By Solution (Hardware, Software, Services), By Technology (Deep Learning, Machine Learning), By End Use, By Region, And Segment Forecasts, 2020–2027*, July 2020, available at <https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-market>; and (2) *IDC Forecasts Strong 12.3% Growth for AI Market in 2020 Amidst Challenging Circumstances*, IDC, August 2020, Available at <https://www.idc.com/getdoc.jsp?containerId=prUS46757920>.

from the selection of one technology or service provider, but through the ability to identify and connect different technologies that maximize the full potential of modern automation technologies. At present, though, software providers are at different stages in developing automation platforms that can harness both RPA and CA technologies. Some enterprises buy best-in-class tools and use their own platforms to integrate them internally.

IBM is a good example. IBM's global IT outsourcing services business combined the Blue Prism RPA software with components of IBM's Watson technology. In one service (Global Technology Services Technology, Innovation & Automation), IBM has well over 1,000 customers, and a range of automation tools, including 200 RPA software licenses provided through its London and Amsterdam offices.

One process that IBM automated by integrating technologies was email ticket triage. IBM's customers have email task IDs they can use to send support requests to IBM. The requests are typically things like "Hey, my printer is not working" or "I've forgotten my password." Any emailed support request needs to be logged into the ticketing system and routed to the correct group for resolution. This task used to be done by humans, but was taken on in 2017 by an RPA software robot. The robot uses its logon ID to log into the ticketing system. It retrieves an email, logs it into the ticketing system and engages the CA tool, which categorizes the issue: "this is a network support request" or "this is a telephony request." The CA tool typically has a high degree of certainty, having been trained on thousands of historical tickets. When confidence is low, the ticket is escalated to a human for routing and the label fed back to the CA tool for learning. The RPA tool closes the loop by identifying the appropriate team and routing the ticket to the correct group.

That example is just one of many. Increasingly, we see CA tools feed into RPA software, which acts as the execution engine, especially in banking, insurance and financial services organizations. A bank, for example, will have an interactive front-end chatbot for dialogue with customers, but it will draw on RPA to get the information it needs to be able to have a more accurate conversation with the customer, for example, about a stolen credit card. In the future, we will see even

more, better integration of RPA and CA software, leading to automation exchange platforms that are increasingly cloud-based. By 2023 the challenge we had identified is how these automation efforts link with all the other digital technologies that might form part of an organization's digital transformation efforts (Willcocks et al., 2023).

Getting Business Value

We, and others, have studied implementations with outcomes ranging from what we describe as “triple-wins” to complete failures. We think it's most useful to focus on success stories because practitioners like to learn from, and hopefully mimic, the achievements of early adopters. But we also studied failures so we could identify the action principles that differentiate outcomes.

Many of our case study organizations achieved triple-wins by achieving value for three types of stakeholder: the enterprise, customers and—most surprising of all—employees. The Associated Press, BNY Mellon, Bouygues, Deakin University, Ericsson, EY Tax Advisory, KPMG, Mars, Nielsen Holdings, Nokia, nPower, SEC Bank, Shop Direct, Standard Bank, Telefónica O2, the VHA, Xchanging and Zurich Insurance are examples of companies that achieved at least one source of value from each category. Aggregating these findings, we have listed the specific benefits and sources of enterprise, customer and employee value that RPA and CA has delivered across our case study companies (see Fig. 9.1).

If this list seems too good to be true, keep in mind that we initially studied known successes, to see if there was any “there” there! Our Telefónica O2 case identified the value to the company in terms of cost savings, the value to customers who received faster services and the value to internal employees, who were released from dreary work to focus on more interesting tasks. The Deakin University case provided details of a triple win for three stakeholders: the university raised its global brand awareness; students gained faster access to critical services; and staff were able to focus on more interesting tasks. Further evidence from other companies is in several other published studies, both by us and other



Fig. 9.1 Triple-Wins of service automation value are evident across multiple RPA and CA implementations

authors (as examples only see Asatiani & Penttinen, 2016; Davenport & Kirby, 2016; Hallikainen et al., 2018; Lacity & Willcocks, 2016; Lacity et al., 2015; Lowes & Cannata, 2017; Schatsky et al., 2016; Watson, 2017; Willcocks et al., 2019).

What do the failures look like? We purposefully started researching failures for our second book, *Robotic Process Automation and Risk Mitigation: The Definitive Guide*. It's always hard to find companies willing to talk about their blunders, though there are quite a few accounts of the limitations as well as the benefits of the automation we are describing.⁵ The software providers and consulting firms we were working with at the time, including Alsbridge (now part of ISG), Blue Prism, Everest Group, HfS, ISG and KPMG, helped us identify failure cases. We studied many enterprises where their RPA implementations had failed to deliver value. Problems included: employee sabotage resulting from inappropriate human resource (HR) policies; functional failures due to poor change management; and implementations that irritated, rather than delighted, customers. Some organizations had localized successes with a few projects, but they never succeeded in scaling and maturing the RPA capability across the enterprise.

What was enlightening to practitioners was to discover that material risks, that is risks that could seriously affect business value, occurred

⁵ Examples of successes and failures can be found in: (1) Davenport (2018), (2) Smith (2019), (3) Broussard (2018), (4) Smith (2018), and (5) Russell (2019).

throughout the automation lifecycle. These were not plug and play technologies; they needed constant management attention. Given our book theme, let's look at sourcing risks more closely (Fig. 9.2).

The main approach we saw was the DIY plus consultancy support option. However, to be truly effective, this has to be a sustained approach to build internal automation capabilities that link with wider digital transformation efforts over a three-year time line. We saw all too many organizations fail to sustain such an investment because of the related challenges that emerged. The result was that even by the early 2020's many automation initiatives had not scaled and become strategically impactful for the organization. That said, the pandemic period did accelerate automation adoption, though often for short-term reasons. One interesting finding was that automation allowed organizations to re-shore work, choosing to go down the automation route internally, rather than outsource on a more traditional basis.

Meanwhile, some organizations chose to utilize their service providers to take over certain work that could be automated. Increasingly, service providers have taken on board utilizing automation themselves, and this has tended to help build their ability to bid for automatable work. In earlier work on sourcing our larger finding was that the low risk approach is to outsource only those activities that are stable, understood and you can write a detailed management contract for. Following this logic, automation is best managed by building up an internal capability first, and outsourcing later when it has become a commodified set of activities (see also this book's preface).

Meanwhile, with the coming of "Robotistan" outsourcing vendors have found themselves from 2015 increasingly in a double bind—firstly not able to depend so heavily on their selling of human labor at low cost, and secondly increasingly having to deliver not traditional but digital services. During the 2017–2020 period a significant minority of clients, we surveyed had reduced reliance on their service providers. The major IT and business process outsourcing (BPO) providers were well aware that their labor arbitrage business model was severely under threat by automation—they feared losing long-time customers who needed fewer services from outsiders after automation. One provider, OpusCapita, made the strategic decision to build up a significant RPA capability

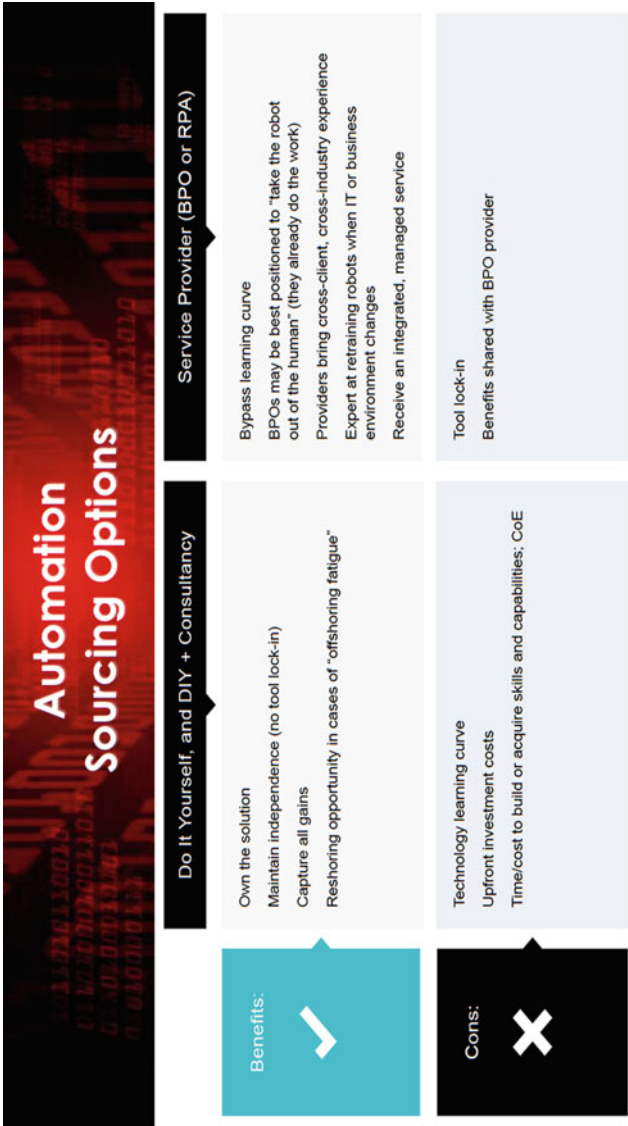


Fig. 9.2 Sourcing options for automation

internally, and then extended RPA services to customers (see Asatiani & Pennttinen, 2016; Hallikainen et al., 2018). Many other outsourcing providers have followed suit since.

Establishing Effective Management Actions

More broadly, we were interested in establishing a full suite of management actions that were leading to higher business value from automation deployment. Our initial rich, cross-sectoral research base for developing what we call “action principles,” covered robotic process and cognitive automation case studies, and is summarized in Table 9.1. (following increasing adoption activity, this has since expanded to over 1,020 RPA and CA cases as at 2023).

What did we find? Do some of the classic practices leading to successful IT project completion apply or, taking those as given, are there distinctive practices critical for RPA/CA success? Overall, we identified 39 action principles that were associated with good outcomes (see Fig. 9.3), and these principles continue to be validated through later research projects.

The action principles approach assumes the following:

Assumption 1: Practitioners are thoughtful agents. Practitioners are assumed to be thoughtful agents capable of action based on free will, power, intelligence, emotion, creativity and self-reflection, but who operate within the affordances and confines of their environments (see Giddens, 1984; Lacity, 2019). Furthermore, practitioners are able to express reasons for and consequences of their actions and the actions of others, *if asked*. As Anthony Giddens wrote: “*To be a human being is to be a purposive agent, who both has reasons for his or her activities and is able, if asked, to elaborate discursively upon those reasons*” (Giddens, 1984). From this assumption, it follows that IS researchers must engage in a dialogue with practitioners to learn about the associations they make between actions and results.

Table 9.1 Client adoption journeys

Company name or pseudonym*	Industry	Head-quarters	Adoption location	Realm	First processes automated
1. Ascension Shared Services	Healthcare	US	BO	RPA	Employee record updates; payments reports
2. Associated Press	Media	US	BO	RPA	Corporate earnings reports
3. Telefónica O2	Telecommunications	UK	BO	RPA	SIM Swaps; pre-calculated credit
4. Xchanging	BPO provider	UK	BO	RPA	Premium advice notices
5. KPMG	Professional Services	US	Innovation Center	CA	Business development; risk assessment; audit
6. Deakin University	Higher Education	Australia	BO	CA	Student queries and engagement
7. SEB Bank	Financial Services	Sweden	IT	CA	IT Services desk; customer services
8. Zurich Insurance	Insurance	Switzerland	BO	CA	Personal injury claims
9. Blue Cross Blue Shield- North Carolina	Healthcare Insurance	US	BO	RPA	Claims processing
10. Biotech*	Biotechnology	Netherlands	BO	RPA	Financial close
11. Building Society*	Financial Services	UK	BO	RPA	Mortgage lending and savings
12. Consulting*	Professional Services	France	BO	RPA	Multiple services
13. Energy*	Natural Gas	Russia	BO	RPA	New customer registration
14. Financial Services*	Financial Services	UK	BO	RPA	Payroll verification

(continued)

Table 9.1 (continued)

Company name or pseudonym*	Industry	Head-quarters	Adoption location	Realm	First processes automated
15. Financial services*	Financial Services	South Africa	BO	RPA/CA	Consumer Credit
16. Healthcare*	Healthcare	UK	BO	RPA	Patient registration
17. IBM Division	Professional and IT services	UK/ Netherlands	IT	RPA/CA	Help desk ticketing
18. Insurance*	Insurance Services	UK	IT	RPA	Pension enrollment
19. Manufacturing*	Consumer Goods	Germany	BO	RPA	Back office processing
20. Utility*	Electric and Gas	Germany	BO	RPA	Meter reading feasibility checks
21. VHA	Healthcare	US	IT	RPA	Web crawls for product descriptions
22. Virgin Trains	Transportation	UK	BO	CA	Incoming customer correspondence

Legend: BO = Business Operations; IT = IT Department

Assumption 2: Action principles are guides, not laws. We assume action principles to be useful guides; we do not consider action principles to be “laws” or even “best practices.” Consistent with David Hume (1739), we are aware of the limitations of claiming blanket generalizations based on empirical evidence. Whereas “best practices” imply that mimicry is always recommended and will always produce similar results, an action principle may not be effective in every context; it is for the thoughtful practitioner to decide. As such, action principles created from early adopters are offered to future adopters for their consideration and guidance. The thoughtful practitioner decides the usefulness of an action principle depending on the objectives the organization is trying to achieve; whether the organization has the absorptive capacity to implement the action principle effectively; and timing—there are better times than others to act. Such future adopters are presumed capable of assessing the applicability of practices garnered from other contexts; it is for the thoughtful practitioner to decide the timing and extent to which action x would likely produce result y within his or her organizational context. A summary of our research approach can be found in Lacity et al. (2021).

Many of our action principles commonly apply to any type of organizational adoption of technology. Examples include: gain C-suite support to legitimate; support and provide adequate resources for the service automation initiative; get stakeholder buy-in; and follow Pareto’s rule by automating the smallest percentage of tasks that account for the greatest volume of transactions. The research shows value in replicating known practices, and reinforces what managers already know about successful technology implementations. Managers need to retain many of their practices in the face of a new technology. But our research did reveal new distinctive practices and some provocative findings (see Fig. 9.3).

Examples of *distinctive* findings in Fig. 9.3 include: under Strategy 1, 3, 5; under Program Management 10; under Process selection 13; Under Tool Selection 16, 18; under Stakeholder Buy-in 22; under Design, Build, Test 25, 26, 27; under Run 28, 29, 31; under Maturity 36, 38.

Strategy 1.You get what you pay for: Focus on long-term value rather than short-term ROI (action) to gain the most value from service automation tools (outcome) 2.Strategy drives investments: Include multiple expected benefits in the justification for service automation investments (action) to achieve the greatest value for the organization, employees and customers (outcome) 3.Consider competitors' reactions (action) to prevent mis-messaging to customers (outcome) 4.Use RPA as forward reconnaissance for CA (action) to gradually build automation skills and to use RPA savings to defer some CA costs (outcome) 5.Strategy envisions longer-term human workforce needs (action) and develops automation and HR plans to gradually meet that vision (outcome)
Sourcing 6. Select the best sourcing option (action) to ensure the success of the implementation (outcome) 7. Incentivize a BPO or tool provider to share the benefits of automation (action) to prevent them from taking all of the savings (outcome)
Program Management 8. Manage RPA as a traditional business case (action) to increase returns on investment (outcome) 9. Manage CA as an innovation program (action) to increase experimentation and to rapidly shift direction (outcome) 10. Manage CA as a learning project (action) to adapt quickly to early lessons learned (outcome) 11. Consider the context (action) to decide which unit is best suited to own the automation program (outcome) 12. Find the "Lewis and Clark" ⁶ program champions (action) who will overcome obstacles to ensure project implementation (outcome)
Process Selection 13. Take the robot out of the human: Use service automation tools to automate mundane tasks (action) to focus employees on more value-adding work (outcome) 14. Aim for triple wins: Develop comprehensive criteria to identify the best processes to automate (action) to achieve the greatest value for the organization, employees and customers (outcome) 15. Fix discoveries about process flaws before deploying service automation (action) to prevent merely performing a bad process more efficiently (outcome)
Tool Selection 16. Use a controlled experiment to assess tools (action) to select the tool that delivers the best financial value (outcome) 17. Don't look for a "Swiss Army Knife": Select a tool that does a few things well (action) to ensure technical success (outcome) 18. Negotiate the optimal level of client-provider transparency pertaining to machine-learning algorithms (action) to ensure a good relationship (outcome) 19. Expect technical challenges as a first mover (action) to minimize disappointments (outcome)
Stakeholder Buy-in 20. Manage up: Gain C-suite support (action) to legitimate, support and provide resources for the service automation initiative (outcome) 21. Manage down: Communicate the intended effect on jobs early in the process (action) to obtain employee buy-in and to prevent panic and sabotage (outcome) 22. Report financial savings in terms of "hours back to the business," rather than as FTE savings (action) to reinforce that automation is used to liberate employees from routine tasks (outcome) 23. Manage expectations: Be transparent with customers that they are interacting with automation tools (action) to maintain high ethical standards and acceptance (outcome)
Design, Build and Test 24. Follow Pareto's rule: Automate the small number of tasks that account for the greatest volume of transactions (action) to deliver the most business value (outcome) 25. Don't underestimate the data challenge (action) required to get CA tools to perform competently (outcome) 26. Find new data sources if "dirty" data cannot be cleaned (action) to get the CA tool to perform competently (outcome) 27. Compare training the automation tool to training a new employee (action) so that stakeholders expect the tools to be as competent as new employees and not as competent as experts (outcome)
Run 28. Redesign employee scorecards so that they are credited with productivity gains contributed by their robot teammates (action) 29. Invite customers to experiment with the automated service, but keep other channels open (action) to ensure good customer service (outcome) 30. Invite customers to provide feedback (action) to help improve the performance of a CA tool (outcome) 31. Keep subject matter experts continually engaged in data curation (action) to keep the automation relevant (outcome) 32. Robots need supervisors: Supervise the learning of the service automation tool (action) to prevent machines from making new decisions without human direction and approval (outcome) 33. Rethink human talent and skills (action) needed for long-term success (outcomes) 34. Assign clear roles of responsibility (action) to keep the automation operational and relevant (outcome)
Maturity 35. Reuse components (action) to scale quickly and to reduce development costs (outcomes) 36. Multi-skill the robots (action) to extract more business value (outcome) 37. Create a center of excellence (action) to disseminate automation technologies across the organization (outcome). 38. Integrate RPA and CA initiatives (action) to deliver end-to-end service automation (action) 39. Continually innovate (action) to deliver value to customers, employees and shareholder (outcome)

Fig. 9.3 Intelligent automation action principles (Source Lacity et al. [2021])⁶

⁶ The Lewis and Clark Expedition from 31 August 1803 to 25 September 1806 was the US expedition to cross the newly acquired western portion of the country. The expedition leaders had to overcome many new challenges.

Jobs and Automation

Some findings were surprising—at least to us. An example of such a *provocative* finding is worth considering in detail. Many people falsely assume that ROI from automation comes from firing employees. While it's true that the primary source of value from service automation is freeing up human labor, the best way to deliver, measure and communicate this value to the enterprise is “hours back to the business.” Think of it as a gift given back to the organization. But what will an organization do with that gift? Most enterprises we studied used the freed-up labor capacity to redeploy people to other tasks within the work unit. Many of these organizations were experiencing high growth, and automation helped them take on more work without hiring proportionally more workers. You can easily imagine how valuable it is to grow efficiently by redeploying existing employees rather than searching for, vetting, onboarding and training new ones.

Let's differentiate “hours back to the business” from what we normally think of as freeing up FTES (full-time equivalents). The two terms are similar, but the messaging is very different. “Hours back to the business” calculations are based on estimating the number of hours it would take if humans still performed the automated tasks. It represents the human capacity that is now free to do different work. Hours back to the business can be converted to FTEs, typically by dividing by 2,000, the average number of hours an employee works per year. For example, EY's US Tax Advisory Business generated 800,000 hours back to the business within 18 months of its RPA implementation. If EY reported that automation saved them 400 FTEs instead of the equivalent 800,000 hours, many may interpret it as: “We no longer need 400 people.” But that sends the wrong message, as the labor savings typically come from automating a portion of people's jobs. In practice, saving 400 FTEs is more likely to come from automating 20% of 2,000 people's jobs.

But surely some automation implementations lead to layoffs? Not necessarily. First, we are talking here about time saved that can be used elsewhere in the organization. Second, the time saved is invariably spread across jobs. It's partial task automation, not job loss. Third, people fail to

understand that there is an enormous amount of extra work being generated every year in organizations—in one study we suggest between 8 and 12% (Willcocks, 2020a, 2021; see also chapter 13). This arises from the exponential explosion in data volumes, increasing audit and regulation requirements, and bureaucracy, as well as the problems information and communications technologies bring with them—cybersecurity as an obvious example. Increasing workloads allied to skills shortages have forced organizations to turn to automation as a coping mechanism, rather than primarily for headcount reduction.

Of course, we did document some layoffs, but—perhaps surprisingly—layoffs have not been as widespread as one might think, especially during the time frame of our study (2015–2023). Other researchers have also documented this perhaps counter-intuitive insight. Davenport and Rananki found that replacing administrative employees was neither a primary objective nor a common outcome of deploying RPA (Davenport, 2018). They found that only 22% of executives sought headcount reduction as the primary objective of “AI.” KPMG found that, of the employees displaced by “AI,” only 14% were let go, with the rest retrained to deal with data, or for AI-related tasks, or to work on a specific process or in an industry domain, or to service new business needs.⁷

More generally, Manyinka and Burghin (2018) identified three simultaneous “AI” impacts: jobs lost, jobs gained and jobs changed. In their mid-point scenario, jobs lost by 2030 could displace 400 million workers, but these losses would be more than compensated for by between 555 and 890 million jobs gained. They also predict that many more jobs would be changed by automation than would be lost.

It’s worth sharing the findings from two surveys we conducted—one in 2017 of 124 senior managers and one in 2018 of 60 service automation adopters. We asked respondents, “What does your organization do with the labor savings generated from automation?” Only 22% of the respondents in the 2017 survey and 15% of the respondents in the 2018

⁷ *Easing the pressure points: The state of intelligent automation*, KPMG International (in collaboration with HFS Research), March 2019, available at <https://assets.kpmg/content/dam/kpmg/xx/pdf/2019/03/easing-pressure-points-the-state-of-intelligent-automation.pdf>.

survey indicated that their organizations laid off employees as a consequence of automation. Other enterprises did reduce headcount, but not through layoffs. Instead, they took a gentler approach to ratcheting down headcount gradually through natural attrition, or by slowing recruitment or offering early retirement.

Our research found that, certainly on 2017/2018 plans, service automation technologies were most commonly used to free up employees from dreary, repetitive work enabling them to focus on more value-adding tasks. Specifically, most enterprises used the freed-up time to redeploy people to other tasks within the work unit or to other work units within the company, for example to reduce the backlog of work or to take on more work without adding more headcount. This was particularly evident in shared service organizations that were under pressure to take on more services without adding more employees. A lot of these organizations were assigning employees to more customer-facing roles and also combining tasks to redefine what constituted a “job.”

There was a further surprising finding—at least to us. Before the field research, we had assumed employees would be threatened by automation. Instead, we found that those who embraced automation⁸ developed highly valued skills, and many of them were either promoted or went on to join or start consulting companies.

The lesson we derived is that human resources (HR) needs to be involved in RPA/CA automation projects if enterprises want to retain their talent. HR can help with new job descriptions, better compensation packages and opportunities for employees to grow. This has been particularly pertinent following the Covid-19 crisis. Following automation, HR may need to redesign employee scorecards. If they don't, an individual's productivity metric might decline after automation because the employee will focus on more complex work as the robots take over the easy tasks. For example, in one case study on healthcare

⁸ Many employees so welcomed automation that they anthropomorphized their software robots by providing them with names, personalities and depictions. We first encountered this back in 2014 at Xchanging (now DXC Technologies), and soon found the phenomenon at other sites. For an account, see Lacity, M. C. and Willcocks, L. (2015) *What Knowledge Workers Stand to Gain from Automation*, Harvard Business Review online, June 19., available at <https://hbr.org/2015/06/what-knowledge-workers-stand-to-gain-from-automation>.

claims processing, on average, each person processed 12 claims per hour before automation. After automation, this fell to about seven claims per hour—simply because they were now working just on the complex claims. The employees were obviously unhappy because it looked as if their productivity had dropped from 12 claims per hour to seven. The healthcare company understood and adjusted expectations and employee compensation.

Automation Strategy Recommendations

Is there other advice emerging from the research, to help those who are starting or progressing through their RPA/CA journeys? For those just starting their journeys, we summarize the action principles with this advice: *think big, start small, institutionalize fast and innovate continually*.

By *thinking big*, we mean that enterprises need to develop an intelligent automation capability to thrive in the twenty-first century. They need to think strategically about automation from the start by focusing on value-driven triple-wins—for the enterprise, its customers and employees. On our 2023 evidence they also need to align automation deployment with the wider digital strategy, and digital transformation efforts (Willcocks et al., 2023).

Enterprises can *start small* with a pilot, but they likely do not need to do a proof-of-concept; the technology has already been proven, particularly for RPA. A pilot should include a business sponsor, IT security, IT operations, compliance teams and HR so that the automation will be designed for production from the start. There are also plenty of competent advisory firms that can help. Starting small quite quickly must link up with the overall digital strategy, and involve integration with the organization's technical architecture blueprint, and its governance arrangements.

Institutionalize fast refers to creating an organizational structure and change-management capability to mature and scale intelligent automation throughout the enterprise. Many companies initially set up a RPA/

CA center of excellence. We have seen such centers evolve into automation, then digital centers of excellence, thus expanding digital governance alongside digital ambition (Willcocks et al., 2023).

Finally, automations need to be managed and continually improved—software robots are like digital employees: they need to be retrained when business rules change and their proficiency can be improved with more feedback. But our most recent research shows a further evolution where automation can lead to process and business innovations. Thus as RPA and CA become an integrated part of the digital platform, so much more is enabled in terms of processes, new services, new customer experiences and ultimately even new business models (Willcocks et al., 2023).

Our advice for progressing an RPA/CA journey is encapsulated in our action principles for running and maturing service automation capabilities. We discuss these and how they were developed in an evidence-based way in Lacity et al., (2021—see also note 16 above). But let us here highlight a few critical ones.

Particularly for CA deployments in customer-facing services, initial performance will not be perfect because, as we have already pointed out, CA tools produce probabilistic outcomes rather than deterministic results. No matter how much organizations tested a CA tool in their sandbox environments, it never precisely predicted how customers would actually interact with the technology after it went live. SEB Bank learned to warn customers that a virtual software assistant would make the first attempt at answering their query and provided quick access to a human if they grew frustrated. Initially at the bank, IPsoft's Amelia handled about 50% of text chat conversations without needing human intervention. By continually analyzing the live customer interactions, SEB Bank trained the tool to perform better over time.

Another key piece of advice is to multi-skill the robots. Initially, many organizations were buying one software license—i.e., one software robot per automation process. This meant that several robots would be idle during different times of the working day. Though employees tend to have specialist skills that cannot be dynamically re-routed to balance out demand fluctuations from across the enterprise, and also require rest periods, a software robot can be programmed to run different tasks and operate 24 hours a day. It took a while before managers realized that

they could program a single software robot to automate several processes, provided they scheduled them at different times. The RPA providers do not like that insight because it results in fewer software licenses being purchased.

Beyond these, smart organizations are now looking to integrate automation into a much bigger digital transformation, and RPA and CA are actually fundamental to such transformations, not a useful “quick-win” appendage. But then automation becomes even more difficult, because digital transformation is a large-scale, long-term, complex process in any sizeable long-standing organization (Wade & Shan, 2020).

Today’s Automation Challenges

As they enter the mid-2020s what are organizations struggling with when it comes to sourcing automation? We have identified several areas where managers and organizations really need help. The first is that they struggle to scale their automation. By the end of 2019, only 13% had scaled and industrialized their RPA deployment and only 12% had an enterprise approach to automation. We looked at this again in late 2020, and while the top software providers had lots of customers, very few of those customers had more than 100 software robots.⁹ Let’s move to late 2022. Looking just at the 80 plus major “connected-RPA” (RPA plus intelligent automation) suppliers, most clients had between 1 and 80 RPA “robots” (licenses). Few (25%) had scaled to 81–100, let alone a higher number. This has been changing in the last year, but reflects scaling, strategic investment and benefits aspiration challenges.

We think this is partly because the cost of getting to the next stage looks steep, though we have evidence that suggests the benefits can be exponential. Second, there are problems with integrating RPA technology with existing or new IT, let alone across the enterprise. A big issue is pre-existing process fragmentation. The challenges of RPA deployment

⁹ In a December 2020 survey by Deloitte of 441 executives, only 13% of organizations had 51 or more robots. See *Automation With Intelligence*, 2020, Deloitte Insights, available at <https://www2.deloitte.com/content/dam/Deloitte/tw/Documents/strategy/tw-Automation-with-intelligence.pdf>.

are compounded where C-suite executives do not see these technologies as strategic, stay too remote from the programs, and frequently under-invest in automation.

The problems of deploying CA technologies are even greater, and each problem is crying out for research attention from a practice-oriented perspective. If you listen to the AI hype, it's very easy to underestimate how slow and challenging actual progress has been to date. We summarize these challenges with an acronym: BOGSHABIF, which stands for brittle, opaque, greedy, shallow, hackable, amoral, biased, invasive and fakeable. It's a pretty rich list of things that practitioners need help with, and humans will be vital in not only designing better CA but partnering with the technology during operational use.

Interestingly, we have seen recently banks and telecom companies driving digital transformation and automation efforts from different starting points and with different executives. As a result, IA initiatives can get becalmed. Moreover, slow progress on digital transformation can delay integrating transformation with the automation agenda. Organizational silos slow deployments of RPA, CA and all the digital technologies. We call the extreme version, which we have encountered, “the 8-siloed organization”—i.e., an organization with multiple silos in processes, skills, managerial mindsets, strategy, structure, data, technology and culture. Scaling and optimizing RPA and CA in such organizations, let alone achieving digital transformation, is indeed a long haul. A lot of research is still needed in this area.

There has also been little research on “born digital” companies and how they use automation technologies. Have they bypassed the “8-silo” problem, and what can more traditional enterprises learn from them? Finally, lots of studies now show that the Covid-19 pandemic and economic crises have accelerated the deployment of automation, but so far, we have found the technologies have been used differently—e.g., to sweat assets or underpin current business performance—or that the crisis might have slowed the automation strategy. Only a few organizations are really investing strategically (as opposed to tactically) in automation—it may be as little as 20% (Willcocks, 2021a, b). This deserves much more study going forward as we are sure the picture will change dramatically into the mid-2020s.

Conclusion: Future Challenges and Research

This chapter has given a subjective researcher account of the formative period for RPA, cognitive and intelligent automation leading into AI. It has provided insights into the evolution of automation as deployed in our work organizations. But the insights, recommendations and action principles must remain provisional and updateable. Going forward the chapter leads to some important research questions:

- How can organizations scale RPA and CA beyond automation islands?
- How can organizations deal with the dramatic skill shifts needed for increasingly digital and virtual businesses enabled by RPA, CA and other automation technologies?
- What human skills are needed in a workforce that will increasingly rely on or work alongside digital workers?
- What role should automation technologies play within broader digital transformation programs?
- What are the critical practices required by digital leaders to ensure strategic automation provides disproportionate business value?
- What emerging technologies will shift innovation priorities for RPA and CA and disrupt workforce and business models even further?
- How can organizational absorptive capacity avoid being outstripped by the pace of technological innovation?

Let's add to this picture. To date, intelligent automation tools have been adopted within the boundaries of the firm, but the next phase is happening in cooperation with ecosystem partners. For example, inter-organizational data sharing, which requires ecosystem-level solutions, is a great example of the next generation of challenges IT leaders must solve (Wixom & Sebastian, 2020). We are now studying how technologies can automate inter-organizational transactions. The solutions often require integrating a variety of technologies, including Internet of Things (IoT), radio frequency identification (RFID), ERP, blockchains, RPA and machine learning. In our recent case studies on ecosystem-level applications, technologies represented between 20 and 30% of the effort required. Up to 80% of the effort was for trading partners to agree on

data and event standards, shared governance models, intellectual property rights and compliance assurance. There's a lot more to investigate in this area, including:

- How can service automation technologies be used to automate inter-organizational transactions?
- Which governance models are effective for shared applications?

Finally, our research method derived the RPA and CA action principles. These principles are based on cumulative evidence thus far, but will need to be revised, extended or retired as organizations gain more experience. We encourage researchers, in particular, to further vet and expand the action principles associated with the maturity phase of adopting intelligent automation.

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